WORKSHEET - NAPLAN Year 9

Number and Algebra
Solving Equations (9)

Teacher: Joanne Rust
Exam Equivalent Time: 10 minutes *(based on NAPLAN allocation of ~1.25 min/mark)*

Questions

1. ☒ Number and Algebra, NAP-48606

The table below shows the results of a rule that changes each ○ into a △.

○	△
2	5
3	9
6	21

Which rule below was used?

- △ = ○ $\times 1 + 3$
- △ = ○ $\times 2 + 1$
- △ = ○ $\times 3 + 3$
- △ = ○ $\times 4 - 3$

2. ☒ Number and Algebra, NAP-72128

Zoe collected 8 baskets of eggs from her chicken coup.

Each basket had the same number of eggs and a total of 96 eggs collected.

Which equation shows the average number of eggs, x, in each basket?

- $x = 96 \times 8$
- $x = 96 - 8$
- $x \div 8 = 96$
- $x \times 8 = 96$

3. ☒ Number and Algebra, NAP-54553

Fabio drove 300 km in $4\frac{1}{2}$ hours.

His average speed for the first 210 km was 70 km per hour.

How long did he take to travel the last 90 km?

[ ] hours

4. ☒ Number and Algebra, NAP-30866

Renee and Leisa are saving money so they can visit their grandmother on a holiday.

Renee has $100 and plans to save $30 each week.

Leisa has $200 and plans to save $10 each week.

After how many weeks will Renee and Leisa have saved the same amount?

[ ] weeks

5. ☒ Number and Algebra, NAP-72158

It takes Kate 15 seconds to place a brochure in an envelope and seal it.

How many minutes will it take her to pack and seal 10 envelopes?

[ ] minutes

6. Number and Algebra, NAP-72162

Ogi owns the most marbles among his friends. He owns

- 4 times the number Chris owns,
- 12 times the number Alison owns and
- 32 times the number Matt owns.

About how many times more marbles does Chris own than Matt?

8 $\quad 28$ $\quad 36$ $\quad 128$

---

7. Number and Algebra, NAP-95710

$\frac{1}{4}$ of Riley's savings equals $\frac{1}{3}$ of Cianna's savings.

Their total savings together is $63.

How much has Cianna saved?

$

---

8. Number and Algebra, NAP-07376

Mal drew the line $y = 4 - x$ on a grid.

He then drew $y = x - 2$ on the same grid.

What are the coordinates of the intersection point of these two lines?

$(-1, 3)$ $\quad (1, 3)$ $\quad (3, -1)$ $\quad (3, 1)$

Worked Solutions

1. Number and Algebra, NAP-48606

$$\triangle = \bigcirc \times 4 - 3$$

---

2. Number and Algebra, NAP-72128

$$\text{Eggs per basket} = \frac{\text{total eggs}}{\text{number of baskets}}$$

$$x = \frac{96}{8}$$

$$\therefore x \times 8 = 96$$

---

3. Number and Algebra, NAP-54553

Time for 1st 210 km

$$= \frac{210}{70}$$

$$= 3 \text{ hours}$$

$\therefore$ Time for last 90 km

$$= 4\frac{1}{2} - 3$$

$$= 1\frac{1}{2} \text{ hours}$$

---

4. Number and Algebra, NAP-30866

After w weeks,

Renee has saved: $100 + 30w$

Leisa has saved: $200 + 10w$

$$\text{If } 100 + 30w = 200 + 10w$$

$$20w = 100$$

$$w = 5$$

$\therefore$ Amounts are equal after 5 weeks.

5. Number and Algebra, NAP-72158

$$\begin{aligned} \text{Time} &= 10 \times 15 \\ &= 150 \text{ seconds} \\ &= 2.5 \text{ minutes} \end{aligned}$$

---

6. Number and Algebra, NAP-72162

Let $O =$ number of Ogi's marbles

$C =$ number of Chris' marbles etc ...

Expressing the information given as equations:

$$\begin{aligned} O &= 4C \\ O &= 32M \\ \therefore 4C &= 32M \\ C &= 8M \end{aligned}$$

$\therefore$ Chris owns 8 times more marbles than Matt.

---

7. Number and Algebra, NAP-95710

$$\frac{1}{4}R = \frac{1}{3}C$$

$$R = \frac{4}{3}C \ \ldots (1)$$

$$R + C = 63$$

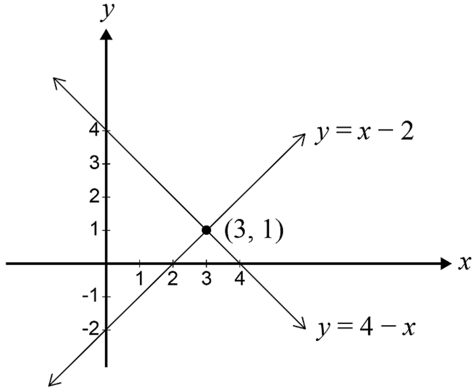
Substitute $R = \frac{4}{3}C$ from (1)

$$\begin{aligned} \frac{4}{3}C + C &= 63 \\ \frac{7}{3}C &= 63 \\ C &= \frac{63 \times 3}{7} \\ &= \$27 \end{aligned}$$

---

8. Number and Algebra, NAP-07376

Strategy one: graph each line and see the intercept.



Strategy two:

Testing each set of coordinates in both equations, consider the 4th option, (3,1).

$1 = 4 - 3$ (satisfies 1st equation)

$1 = 3 - 2$ (satisfies 2nd equation)

$\Rightarrow$ $(3, 1)$ is the intersection point.
